

MAT11-5

Descriptive Statistics & Data Reasoning

Albert School — 22 hours

Preface

You arrive in this course already fluent in two languages. From **Foundations** (MAT11-1) you know functions, monotone transformations, derivatives, and the discipline of checking that a number is the right size and in the right units. From **Linear Algebra I** (MAT11-3) you know vectors, dot products, norms, angles, and projection. This course is where those two languages start describing the real world: messy, finite, noisy datasets.

The temptation in statistics is to treat formulas as buttons — press “mean,” press “correlation,” read the output. That is dashboard production, and it is not what we are after. Every quantity here is a **summary**: it throws away information on purpose, and the whole skill is knowing what was thrown away and whether the discarded part mattered. A mean hides skew. A correlation hides nonlinearity. An aggregate hides Simpson’s paradox. Our objective is mathematically disciplined interpretation — computing the number, yes, but always asking what it conceals.

A recurring habit will serve you throughout: **before stating a formula, look at a concrete dataset and ask what answer you want; then check the formula gives it.** We follow that habit from the first page.

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1 Central tendency

A first dataset. Seven employees at a small startup earn, in thousands of euros per year,

30, 32, 35, 38, 41, 44, 250.

The last figure is the founder. Before any formula: what is a “typical” salary here? Most people would say “the high thirties.” Keep that intuition; we will see which formula respects it and which betrays it.

For a dataset $x_1, \dots, x_n$ we use three summaries of where the data sit.

Definition 1 (Mean, median, mode)

- The **(arithmetic) mean** is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

- The **median** is the middle value once the data are sorted; if n is even it is the average of the two middle values.
- The **mode** is the most frequent value (there may be several, or none useful for continuous data).

Worked example — The three summaries on the salary data. The data are already sorted, $n = 7$.

- Mean: $\frac{30+32+35+38+41+44+250}{7} = \frac{470}{7} \approx 67.1$.
- Median: the 4th of seven values, 38.
- Mode: every value appears once, so the mode is uninformative here.

The mean, 67.1, is a salary **no one earns and exceeds six of the seven people**. The median, 38, matches the intuition we wrote down. A single extreme value — the founder — dragged the mean far from the bulk of the data, while the median did not budge.

This example contains the whole lesson of the chapter.

Which measure, when?

- The **mean** uses every value and is the right summary for roughly symmetric data without extreme outliers. It is the unique “balance point” of the data.
- The **median** ignores how far the extremes are, only how many lie on each side, so it is **robust**: appropriate for skewed data or data with outliers (incomes, house prices, response times).
- The **mode** is the natural summary for categorical data (“most common product sold”) or to flag the most likely value of a discrete variable.

1.1 Skew and the order of the mean and median

The relationship between the mean and median is a fast diagnostic for the **shape** of a distribution.

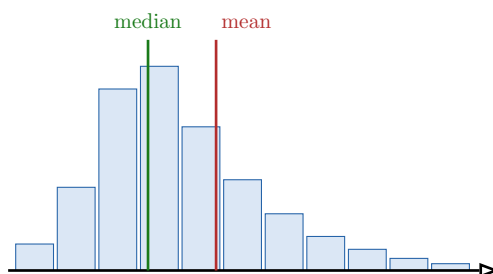


Figure 1: A right-skewed (positively skewed) distribution: a long tail to the right. The mean is pulled toward the tail, so it sits **above** the median. On a left-skewed distribution the picture mirrors and the mean lies **below** the median; on a symmetric distribution the two coincide.

- Right-skewed (long right tail): $\bar{x} > \text{median}$.

- Left-skewed (long left tail): $\bar{x} < \text{median}$.
- Symmetric: $\bar{x} \approx \text{median}$.

Common pitfall The mean is not “the typical value.” On skewed data it can be larger than almost every observation, as on the salary data. Reporting only the mean of incomes, house prices, or city sizes is one of the most common honest-looking ways to mislead. When in doubt, report both and look at the shape.

2 Dispersion

Knowing the centre is half the story; we also need to know how **spread out** the data are. Two classes can both average $\frac{12}{20}$ on an exam while one has everyone near 12 and the other splits between 4 and 20.

Definition 2 (Variance and standard deviation) For data $x_1, \dots, x_n$ with mean $\bar{x}$,

$$s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \quad (\text{variance}), \quad s = \sqrt{s^2} \quad (\text{standard deviation}).$$

The variance is the mean squared distance from the centre; the standard deviation s rescales it back into the **same units as the data**.

Remark Why square the deviations instead of just averaging $|x_i - \bar{x}|$? Squaring makes the algebra clean (it connects to dot products and projection, as we will see in the LA bridge) and makes $\bar{x}$ the value that **minimises** the total — the mean is the point of least squared distance. The price is sensitivity to outliers, since a far-away point contributes its distance **squared**.

Worked example — Two classes. Class A: 10, 11, 12, 13, 14. Class B: 4, 8, 12, 16, 20. Both have mean 12. For A the squared deviations are 4, 1, 0, 1, 4, so $s_A^2 = \frac{10}{5} = 2$ and $s_A \approx 1.41$. For B they are 64, 16, 0, 16, 64, so $s_B^2 = \frac{160}{5} = 32$ and $s_B \approx 5.66$. Same centre, four times the spread.

Remark You may meet a variant with $\frac{1}{n-1}$ instead of $\frac{1}{n}$ — the **sample** variance, which corrects a bias when estimating the spread of a larger population from a sample. For pure description of the data in hand we use $\frac{1}{n}$; the distinction becomes important in inferential statistics and we flag it here only so the two formulas do not surprise you.

2.1 A robust alternative: quartiles and the IQR

The standard deviation inherits the mean’s fragility: one outlier inflates it. The robust counterpart uses **order** rather than magnitude.

Definition 3 (Quartiles and interquartile range) Sort the data. The **first quartile** Q_1 is the 25th percentile (a quarter of the data lie below it), the **median** is the 50th, and the **third quartile** Q_3 is the 75th. The **interquartile range** is

$$\text{IQR} = Q_3 - Q_1,$$

the width of the middle 50% of the data.

Because Q_1 and Q_3 only care about position, moving the largest value from 250 to 250000 leaves the IQR unchanged — exactly the robustness the standard deviation lacks.

2.2 Box plots

A **box plot** draws the **five-number summary** — minimum, Q_1 , median, Q_3 , maximum — at a glance.

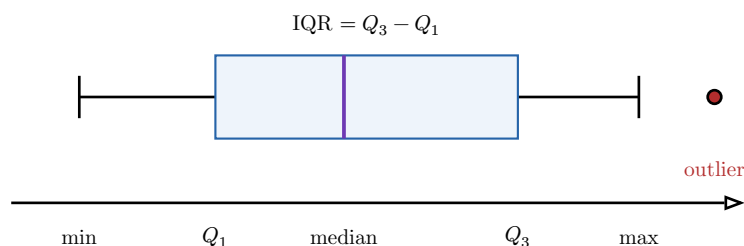


Figure 2: Anatomy of a box plot. The box spans the middle 50% (from Q_1 to Q_3), the inner line is the median, and the whiskers reach to the most extreme points still within $1.5 \times \text{IQR}$ of the box. Points beyond the fences $Q_1 - 1.5 \times \text{IQR}$ and $Q_3 + 1.5 \times \text{IQR}$ are drawn individually as candidate outliers.

Definition 4 (The 1.5 IQR rule for outliers) A point is flagged as an outlier if it lies below $Q_1 - 1.5 \times \text{IQR}$ or above $Q_3 + 1.5 \times \text{IQR}$. These cutoffs are the **fences**; the whiskers stop at the most extreme data still inside them.

Common pitfall Dispersion has no absolute meaning — it must be read against the data-generating process. A standard deviation of 0.5 mm is alarming for a machined part tolerated to 0.1 mm but negligible for hand-cut timber. Always interpret spread **relative** to how variable the process should be.

3 Bivariate statistics

So far, one variable at a time. The richer questions are about **pairs**: do study hours and exam scores move together? Does price move with demand?

A paired dataset. Five students report (hours studied, exam score):

$$(1, 52), \quad (2, 58), \quad (3, 64), \quad (4, 78), \quad (5, 83).$$

As hours rise, scores rise. We want a single number for “how much, and in which direction, do these move together.”

Definition 5 (Covariance) For pairs $(x_1, y_1), \dots, (x_n, y_n)$ with means $\bar{x}, \bar{y}$,

$$\text{cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

The idea is simple once you read the sign of each term. Centre both variables. If x_i and y_i are **both** above their means, or **both** below, the product $(x_i - \bar{x})(y_i - \bar{y})$ is positive; if one is above while the other is below, it is negative. Covariance averages these signed agreements.

Worked example — Covariance is hard to read alone. For the study data, $\bar{x} = 3$, $\bar{y} = 67$. The centred products are $(-2)(-15)$, $(-1)(-9)$, 0 , $(1)(11)$, $(2)(16) = 30, 9, 0, 11, 32$, summing to 82, so $\text{cov} = \frac{82}{5} = 16.4$. Positive — good, they move together. But is 16.4 “a lot”? Switch the score to a 0–1 scale and the covariance shrinks by 100; switch hours to minutes and it grows by 60. The number depends on the units, so its **magnitude** is meaningless on its own.

This unit-dependence is exactly what correlation fixes.

Definition 6 (Pearson correlation)

$$r = \frac{\text{cov}(x, y)}{s_x s_y}, \quad -1 \leq r \leq 1,$$

where s_x, s_y are the standard deviations. Dividing by the spreads cancels the units, so r is **dimensionless**.

Reading r .

- **Sign** — direction: $r > 0$ rises together, $r < 0$ one rises as the other falls.
- **Magnitude** — strength of the **linear** relationship: $r = \pm 1$ is a perfect straight line, $r \approx 0$ no linear relationship.
- r is unaffected by the units or by an increasing linear rescaling of either variable.

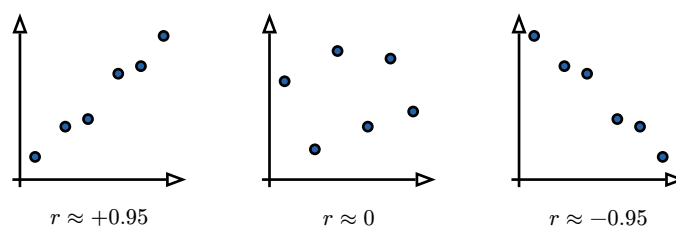


Figure 3: Scatter plots and their correlations. Left: strong positive linear trend. Middle: no linear trend. Right: strong negative trend. A scatter plot shows the **form**, **direction**, and **strength** of a relationship at a glance — always draw it before trusting a single number.

Common pitfall $r = 0$ does not mean “no relationship,” only “no *linear* relationship.” Points lying exactly on the parabola $y = x^2$ symmetric about 0 have $r = 0$, yet y is perfectly determined by x . The scatter plot reveals what r cannot. This is why we visualize before we summarize.

3.1 Correlation is not causation

Common pitfall Correlation does not imply causation. A non-zero r between two variables can arise without either causing the other, through:

- a **common cause** (a confounding variable) — ice-cream sales and drowning deaths both rise with summer heat;
- **reverse causation** — you assumed $X \rightarrow Y$ but in fact $Y \rightarrow X$;
- **coincidence** — with enough variables, some pairs correlate by chance.

Establishing causation needs more than data shape: a controlled experiment, or a careful argument ruling out confounders.

4 Statistical biases

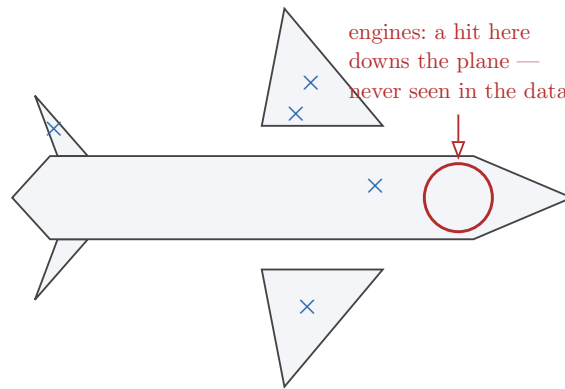
A **bias** is a systematic distortion that pushes conclusions away from the truth — systematic, not random, so collecting more data the same flawed way does not help; it sharpens the wrong answer.

Definition 7 (Four common biases)

- **Sampling bias** — the sample is not representative of the population it is meant to describe.
- **Selection bias** — the rule deciding which units enter the data is itself related to the outcome being studied.
- **Survivorship bias** — only the units that “survived” some process are observed; the casualties are invisible.
- **Confirmation bias** — evidence is sought or read so as to confirm a prior belief.

Worked example — Each bias in the wild.

- **Sampling:** a 1936 US election poll mailed to car and telephone owners predicted the wrong winner — in the Depression, owning a car signalled wealth, so the sample leaned one way.
- **Selection:** judging a drug by averaging only patients who **completed** the trial hides those who quit because it was failing them.
- **Survivorship:** in WWII, returning bombers were riddled with bullet holes on the wings and tail. The instinct was to armour those spots. Abraham Wald saw the inversion: the planes hit in the **engines** never returned to be counted, so armour belonged exactly where the survivors had **no** holes.
- **Confirmation:** an analyst sure a product will succeed records the glowing reviews and explains away the bad ones as “not the target user.”



blue x: where *surviving* planes were hit

Figure 4: Survivorship bias, the bomber case. The damage you can see (blue) marks where a plane can be hit and still return. The danger is the region with **no** visible damage — hits there removed the plane from the sample entirely.

4.1 Simpson's paradox

The subtlest distortion of all: a trend can hold in **every** subgroup yet **reverse** when the groups are pooled.

Worked example — Admissions reversal. A university is accused of bias against women. Aggregated, men are admitted at a higher rate. Yet department by department, women are admitted at an **equal or higher** rate. How? Women applied disproportionately to the most competitive department, which admits few applicants of **either** sex. The department, not the sex, drives the low rate; pooling across departments with very different selectivity creates the illusion.

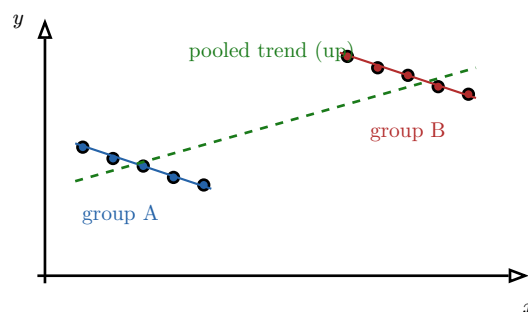


Figure 5: Simpson's paradox. Within each group the trend slopes **down** (solid lines), but because group B sits at high x and low y , the pooled cloud slopes **up** (dashed). The lurking variable — group membership — is distributed unevenly across x . Diagnosing the paradox means computing the relationship both within subgroups and in aggregate, then naming the variable whose uneven distribution flips it.

5 Data visualization

A chart is a summary too, and the same discipline applies: each chart type answers one kind of question, and using the wrong one hides the answer or invents a fake one.

Choosing the chart from the data type.

- **Bar chart** — one numeric value compared across **categories**.
- **Histogram** — the **distribution** of one numeric variable, by binning it into intervals.

- **Scatter plot** — the **relationship** between two numeric variables.
- **Box plot** — **compare distributions** (spread, median, outliers) across several groups.
- **Heatmap** — a **matrix** of values encoded by colour, e.g. a correlation matrix.

Common pitfall A histogram and a bar chart look alike but answer different questions. A bar chart's x -axis is **categorical** (gaps between bars are meaningful); a histogram's x -axis is **numeric and continuous** (bars touch, and bin width matters). Mislabelling one as the other is a common beginner error.

5.1 Misleading visualizations

Worked example — The truncated axis. Two products sell 102 and 108 units. Plotted with the y -axis starting at 100, the second bar looks three times taller — a 6% difference dressed up as a landslide. Start the axis at 0 and the bars are nearly equal.

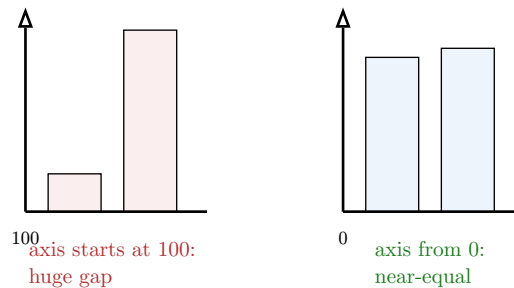


Figure 6: The same two numbers, 102 and 108. Truncating the bar-chart axis (left) turns a 6% difference into an apparent landslide; an honest axis from zero (right) shows them as they are.

A checklist of common chart deceptions.

- A bar-chart y -axis that does not start at zero exaggerates differences.
- A truncated, inverted, or inconsistent axis scale distorts trends.
- Unequal histogram bin widths misrepresent the distribution's shape.
- Encoding a one-dimensional quantity by **area** or **volume** exaggerates ratios (double the radius, quadruple the apparent size).
- Cherry-picking a time window, omitting data, or pairing two unrelated series on a dual axis to suggest a link.

6 Exploratory data analysis

Exploratory data analysis (EDA) is the disciplined first pass over a dataset, before any modelling — the stage where you let the data surprise you.

Definition 8 (The EDA workflow)

1. **State a hypothesis** or question — what do you expect to find, and why?
2. **Compute summary statistics** — central tendency and dispersion for each variable.
3. **Visualize distributions** — histograms and box plots for single variables, scatter plots for pairs.
4. **Detect outliers** — via the $1.5 \times \text{IQR}$ rule or by inspecting box plots, then **investigate** rather than blindly delete them.

5. **Examine missing data** — count missing values per variable and look for patterns: are they missing at random, or systematically (e.g. high earners declining to report income)?

The distribution's **shape** — symmetric or skewed, unimodal or multimodal — feeds back into step 2: it tells you whether the mean or the median, the standard deviation or the IQR, is the honest summary. EDA is a loop, not a checklist.

Common pitfall An outlier is a **question**, not a verdict. It may be a typo (a height of 1075 cm), a genuine extreme (the founder's salary), or the most interesting data point you have (a fraud, a breakthrough). Deleting outliers to “clean” the data, before understanding them, is how real signal gets thrown away.

Common pitfall Missing data is rarely missing at random. If the missingness itself depends on the value — sick patients skip follow-ups, dissatisfied customers ignore surveys — then dropping those rows reintroduces **selection bias**. Always check **why** values are missing before deciding how to handle them.

7 Bridge to Foundations: scale and transformations

Here the **Foundations** toolkit pays off. Summary statistics live on a **scale**, and the monotone transformations you studied there move them in predictable ways.

7.1 Linear rescaling

Proposition 9 (Effect of an affine change of units) If $y_i = ax_i + b$ for constants a, b , then

$$\bar{y} = a\bar{x} + b, \quad s_y = |a|s_x, \quad s_y^2 = a^2s_x^2.$$

Proof. Linearity of the sum gives $\bar{y} = \frac{1}{n} \sum (ax_i + b) = a\bar{x} + b$. Then $y_i - \bar{y} = a(x_i - \bar{x})$, so each squared deviation scales by a^2 , and averaging gives $s_y^2 = a^2s_x^2$; the additive shift b cancels and never touches the spread. $\square$

Worked example — Celsius to Fahrenheit. With $F = 1.8C + 32$: the mean temperature converts the same way, the standard deviation multiplies by 1.8, and the variance by $1.8^2 = 3.24$. The shift $+32$ moves the centre but leaves the spread untouched — exactly as the proposition predicts.

7.2 The logarithm: taming skew and growth

A **logarithmic transformation** $y = \log x$ (for positive data) compresses a long right tail toward symmetry and turns **multiplicative** structure into **additive** structure: a tenfold jump becomes a constant step. It is the right move for data spanning several orders of magnitude — incomes, city populations, file sizes.

It also reads **growth**. A **growth rate** is the relative change $\frac{x_{t+1} - x_t}{x_t}$. Under constant growth, x_t is geometric, so $\log x_t$ is **linear in t** — constant exponential growth becomes a straight line on a log scale, and its slope is the growth rate. This is precisely the “monotone transformation linearises” idea from Foundations, now in the service of data.

Common pitfall Pearson's r is invariant under an **increasing linear** rescaling of each variable (the $ax + b$ above), but **not** under a nonlinear one. Taking log of a variable changes the linear structure of the relationship, so it changes r . Decide whether you want to summarize the raw relationship or the transformed one — they are different questions.

7.3 The z-score: making variables comparable

Definition 10 (Standardization) The **z-score** of an observation is

$$z_i = \frac{x_i - \bar{x}}{s_x}.$$

The standardized data have mean 0 and standard deviation 1.

A z-score says how many standard deviations a value lies from its mean, stripping away units entirely. That is what lets you compare an exam scored out of 20 with one scored out of 100: convert both to z-scores and a +1.5 is a +1.5 on either. (Note this is the affine map with $a = 1/s_x$, $b = -\bar{x}/s_x$, so the previous proposition guarantees the resulting mean 0 and spread 1.)

8 Bridge to Linear Algebra I: covariance as geometry

The single most clarifying idea in this course: **covariance is a dot product**, and **correlation is the cosine of an angle**. Everything from MAT11-3 transfers.

Centre each variable and stack the centred values into vectors in $\mathbb{R}^n$:

$$\tilde{\mathbf{x}} = (x_1 - \bar{x}, \dots, x_n - \bar{x}), \quad \tilde{\mathbf{y}} = (y_1 - \bar{y}, \dots, y_n - \bar{y}).$$

Proposition 11 (The statistics–geometry dictionary)

$$\text{cov}(x, y) = \frac{1}{n}(\tilde{\mathbf{x}} \cdot \tilde{\mathbf{y}}), \quad s_x = \frac{1}{\sqrt{n}}\|\tilde{\mathbf{x}}\|, \quad r = \frac{\tilde{\mathbf{x}} \cdot \tilde{\mathbf{y}}}{\|\tilde{\mathbf{x}}\|\|\tilde{\mathbf{y}}\|} = \cos \theta,$$

where θ is the angle between the centred vectors.

So covariance **is** the (scaled) dot product — the algebraic measure of co-movement — and correlation is the cosine of the angle between the two centred data vectors. The bound $-1 \leq r \leq 1$ is now obvious: it is just $|\cos \theta| \leq 1$. And the extremes have a geometric meaning:

- $r = +1$: $\theta = 0$, the vectors point the **same way** (parallel).
- $r = -1$: $\theta = \pi$, they point **opposite** ways (anti-parallel).
- $r = 0$: $\theta = \pi/2$, the centred vectors are **orthogonal**.

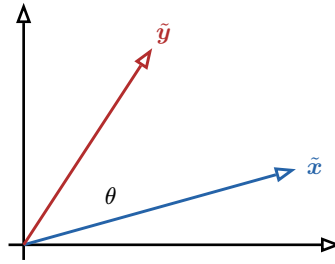


Figure 7: The two centred data vectors. Their correlation is $\cos \theta$: a small angle (vectors nearly aligned) means r near $+1$, a right angle means $r = 0$, and an obtuse angle means $r < 0$.

8.1 Linear trend as projection

Fitting a linear trend $y \approx ax + b$ to a scatter plot is, geometrically, the **projection** you met in Linear Algebra I. Centre the data; then fitting the best line through the origin amounts to projecting the centred response vector $\tilde{\mathbf{y}}$ onto the direction of the centred predictor $\tilde{\mathbf{x}}$.

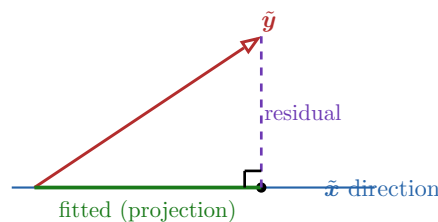


Figure 8: Linear trend as projection. The fitted values are the orthogonal projection of $\tilde{\mathbf{y}}$ onto the $\tilde{\mathbf{x}}$ direction (green); the residuals are the leftover orthogonal component (purple). Least-squares fitting **is** projection — minimising the residual length is the same as dropping a perpendicular, exactly the construction from Linear Algebra I.

This is why “least squares” and “orthogonal projection” are the same idea: the best linear fit is the one whose residual vector is perpendicular to the predictor.

9 Python: a reproducible analysis

The point of the tools is **reproducibility**: anyone should be able to rerun your notebook and obtain your figures and numbers. We use `pandas` for manipulation and `matplotlib/seaborn` for visualization.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("data.csv")

# --- Cleaning: understand missingness before dropping anything ---
df.isna().sum()          # how many missing per column?
df = df.dropna(subset=["x", "y"])  # drop only after understanding why

# --- Summary statistics ---
df.describe()            # mean, std, min, quartiles, max
df["x"].median()
df[["x", "y"]].cov()      # covariance matrix
df[["x", "y"]].corr()     # Pearson correlation matrix

# --- Transformations (Foundations bridge) ---
df["log_x"] = np.log(df["x"])  # tame right skew
```

```

df["z_x"] = (df["x"] - df["x"].mean()) / df["x"].std() # standardize

# --- Visualization: one chart type per question ---
sns.histplot(df["x"]) # distribution of one variable
sns.boxplot(x=df["x"]) # spread + outliers
sns.scatterplot(data=df, x="x", y="y") # relationship between two
sns.heatmap(df.corr(), annot=True) # correlation matrix
plt.show()

```

A reproducible analysis records the **cleaning steps**, the **visualizations**, and — crucially — a **written interpretation** that states at least one **limitation**: a bias the sample might carry, a confounder you cannot rule out, a transformation that changes what the numbers mean. Stating the limitation is not a weakness of the analysis; it is the mark of a disciplined one.

Exercises

1. Compute the mean, median, and mode of a dataset of your choice, then alter a single value to a large outlier and recompute all three. Explain which moved, which did not, and which measure you would report for skewed data.
2. For the salary data 30, 32, 35, 38, 41, 44, 250, decide from the mean and median alone whether the distribution is left- or right-skewed, and confirm with a sketch.
3. Compute the variance, standard deviation, and IQR of a dataset, then interpret each **relative to the data-generating process** (is this spread large or small for what produced the data?).
4. Draw a box plot from a five-number summary and mark the fences and any outliers using the $1.5 \times \text{IQR}$ rule.
5. Compute the covariance and Pearson correlation of two variables. Then change the units of one variable and recompute both; explain why one number changed and the other did not.
6. Construct a dataset of points on $y = x^2$ symmetric about 0 and show $r = 0$ despite a perfect functional relationship. What does this tell you about what r measures?
7. Give an example of two correlated variables that are not causally related, name the common cause, and explain which of confounding, reverse causation, or coincidence is at work.
8. For a study you choose, identify whether sampling, selection, survivorship, or confirmation bias is present, and explain concretely how it distorts the conclusion.
9. Construct or simulate a dataset exhibiting Simpson's paradox. Compute the relationship within each subgroup and in aggregate, and identify the variable whose uneven distribution causes the reversal.
10. For several given data types, choose an appropriate chart and justify it. Then take a misleading chart (e.g. a truncated-axis bar chart) and redraw it honestly.
11. Conduct a full EDA workflow on a small dataset: state a hypothesis, compute summary statistics, visualize the distributions, detect outliers, and check the pattern of missing data.
12. Apply a logarithmic transformation and a z-score transformation to a dataset. Explain how each changes the mean and standard deviation, and why the log transform also changes the correlation with a second variable.
13. Centre two variables, form the centred vectors, and verify numerically that r equals the cosine of the angle between them. Explain the projection interpretation of fitting a linear trend.
14. Produce a reproducible `pandas + matplotlib/seaborn` analysis with data cleaning, at least two appropriate visualizations, and a written interpretation that states one explicit limitation of the data or method.